Introduction to ML - Assignment 1 - Solutions

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1 Correlation and Independence

1.1 A. Prove that independent random variables are uncorrelated

To prove that two independent random variables X and Y are uncorrelated, we show that:

$$E[XY] = E[X]E[Y].$$

Since X and Y are independent, their joint probability density function factorizes:

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x,y) dx dy.$$

Using independence:

$$E[XY] = \int_{-\infty}^{\infty} x f_X(x) dx \cdot \int_{-\infty}^{\infty} y f_Y(y) dy = E[X]E[Y].$$

Thus, $\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y] = 0$, proving they are uncorrelated.

1.2 B. Example of uncorrelated but not independent variables

Let X be uniform on $[-1,1]$ and define $Y = X^2$. We calculate:

$$E[X] = \int_{-1}^{1} x \frac{1}{2} dx = 0.$$

Similarly,

$$E[XY] = E[X^3] = \int_{-1}^{1} x^3 \frac{1}{2} dx = 0.$$

Since $E[X]E[Y] = 0 \cdot E[X^2] = 0$, X and Y are uncorrelated. However, they are not independent since Y is completely determined by X.

1.3 C. Example of three dependent variables with independent pairs

Let X, Y be independent fair coin flips (± 1 with equal probability), and define $Z = XY$. We check independence:

- $P(X = 1, Y = 1) = \frac{1}{4}$, $P(X = 1)P(Y = 1) = \frac{1}{4}$, so X and Y are independent.
- $P(X = 1, Z = 1) = P(X = 1, Y = 1) + P(X = 1, Y = -1) = \frac{1}{2}$, same as $P(X = 1)P(Z = 1)$, so X and Z are independent.
- Similar calculations show Y and Z are independent.
- But knowing X and Y determines Z, proving dependence.

1.4 D. Prove that the empirical correlation matrix C is positive semi-definite

Given:

$$C = \sum_{i=1}^{n} x_i x_i^\top,$$

for any vector $z \in \mathbb{R}^d$:

$$z^\top C z = \sum_{i=1}^{n} (x_i^\top z)^2 \geq 0.$$

Since it is a sum of squares, C is positive semi-definite.

2 Multivariate Gaussians

2.1 A. Probability of being within one standard deviation

For $X \sim N(0, 1)$, we compute:

$$P(-1 \leq X \leq 1) = \Phi(1) - \Phi(-1).$$

From normal tables:

$$\Phi(1) \approx 0.8413, \quad \Phi(-1) \approx 0.1587,$$

thus,

$$P(-1 \leq X \leq 1) = 0.8413 - 0.1587 = 0.6826.$$

2.2 B. Probability decaying to zero as $d \to \infty$ (alternative approach)

Let $X \sim N(0, I_d)$, meaning each coordinate $X_i \sim N(0,1)$ independently. Consider the probability that a sample lies within a unit ball centered at the origin:

$$B_d = \left\{ x \in \mathbb{R}^d : \|x\| \leq 1 \right\}.$$

Now, for the sample $x = (x_1, x_2, ..., x_d)$ to lie within the unit ball:

$$\sum_{i=1}^{d} x_i^2 \leq 1.$$

Since each x_i is standard normal, the average squared length of the vector increases with d. Most of the probability mass is concentrated in a thin shell at a distance $\sqrt{d}$ from the origin.

We upper bound the probability by noting that:

- The unit ball B_d is contained inside a cube of side length 2: $[-1, 1]^d$.
- Thus,
$$P(x \in B_d) \leq P(x \in [-1,1]^d) = \prod_{i=1}^{d} P(x_i \in [-1, 1]).$$
- Since $P(x_i \in [-1, 1]) = 0.6826$ (as computed earlier),
$$P(x \in [-1,1]^d) = (0.6826)^d.$$
- As $d \to \infty$, $(0.6826)^d \to 0$ exponentially fast.

Therefore, the probability that a sample lies within the unit ball decays to zero as $d \to \infty$.

3 Linear Transformation of the Covariance Matrix

Given $y = w^\top X$:

$$\text{Var}(y) = E[(w^\top X - w^\top E[X])^2].$$

Expanding using linearity:

$$\text{Var}(y) = E[w^\top (X - E[X])(X - E[X])^\top w].$$

Factoring $w^\top$ and w:

$$\text{Var}(y) = w^\top E[(X - E[X])(X - E[X])^\top] w = w^\top C w.$$

4 Eigen Decomposition

4.1 A. Prove that A and $A^\top$ share the same eigenvalues

If $Av = \lambda v$, then taking the transpose:

$$(Av)^\top = (\lambda v)^\top \Rightarrow v^\top A^\top = \lambda v^\top.$$

Thus, $A^\top$ has the same eigenvalues.

4.2 B. Determinant as product of eigenvalues

Using the determinant property of diagonalizable matrices:

$$A = PDP^{-1} \Rightarrow \det(A) = \det(P)\det(D)\det(P^{-1}).$$

Since $\det(P^{-1}) = 1/\det(P)$, we get:

$$\det(A) = \det(D) = \prod_{i=1}^{n} \lambda_i.$$

5 Gradients

5.1 A. Piecewise Function

$$f(w, x) = \begin{cases} 0, & w^\top x \geq 1 \\ 1 - w^\top x, & \text{otherwise} \end{cases}$$

Gradient:

$$\nabla_x f(x) = \begin{cases} -w, & w^\top x < 1 \\ \text{undefined}, & w^\top x = 1 \end{cases}$$

5.2 B. Sigmoid Function

$$f(x) = \frac{1}{1 + e^{-w^\top x}}$$

Derivative:

$$\nabla_x f(x) = \frac{d}{dx}\left(\frac{1}{1 + e^{-w^\top x}}\right) = \frac{e^{-w^\top x} w}{(1 + e^{-w^\top x})^2}.$$

Rewriting using $f(x)$:

$$\nabla_x f(x) = f(x)(1 - f(x))w.$$